

Eclipse 2025.4 — Faults & Non-Neighbour Connections

1-to-1 mathematical model with manual equation cross-references

Library: my-axon/libraries/eclipse-faults-nncs

Built 2026-05-23 from Eclipse 2025.4 Technical Description and Reference Manual

1 Scope and notation

This document gives the formal equations Eclipse uses to compute transmissibilities for:

- Block-centred grids (OLDTRAN, OLDTRANR)
- Corner-point grids (NEWTRAN, the default with COORD/ZCORN)
- Half-cell directional permeability (HALFTRAN, E100)
- Radial grids
- Non-Neighbour Connections (NNCs) — both auto-generated and user-defined

The *dual-porosity* matrix-fracture transmissibility (*TD Eq. 2.54*) lives in the sibling document at [eclipse-dual-porosity/notes/math-model.pdf](#).

Symbols.

CDARCY Darcy unit conversion constant; 0.00112712 (FIELD), 0.00852702 (METRIC)

PERMX_{*i*}, PERMY_{*i*}, PERMZ_{*i*} permeabilities in cell *i*

RNTG_{*i*} net-to-gross ratio in cell *i*

TMLT composite transmissibility multiplier = MULTX · MULTY · MULTZ · MULTFLT · MULTREGT · ...

DIPC dip correction (for OLDTRAN family)

DX_{*i*}, DY_{*i*}, DZ_{*i*} in OLDTRAN context: cell side lengths; in NEWTRAN context: vector components from cell centre to face centre

A_{*x*}, A_{*y*}, A_{*z*} X-, Y-, Z-projections of the mutual interface area between two cells

T_{*i*}, T_{*j*} half-cell transmissibilities (used in harmonic mean)

2 Block-centred transmissibility (OLDTRAN)

Used by default with DX/DY/DZ/TOPS input.

X-direction transmissibility

$$\text{TRANX}_i = \frac{\text{CDARCY} \cdot \text{TMLTX}_i \cdot A \cdot \text{DIPC}}{B} \quad (\text{TD 2.1})$$

with the interface area

$$A = \frac{DX_i \cdot DY_j \cdot DZ_j \cdot \text{RNTG}_j + DX_j \cdot DY_i \cdot DZ_i \cdot \text{RNTG}_i}{DX_i + DX_j} \quad (\text{TD 2.2})$$

the flow-resistance sum

$$B = \frac{\frac{DX_i}{\text{PERMX}_i} + \frac{DX_j}{\text{PERMX}_j}}{2} \quad (\text{TD 2.3})$$

and the dip correction

$$\text{DIPC} = \frac{\text{DHS}}{\text{DHS} + \text{DVS}} \quad (\text{TD 2.4})$$

with

$$\text{DHS} = \left(\frac{DX_i + DX_j}{2} \right)^2 \quad \text{DVS} = (\text{DEPTH}_j - \text{DEPTH}_i)^2$$

Y-direction transmissibility

Analogous to (TD 2.1) with $X \rightarrow Y$ in all subscripts.

Z-direction transmissibility

$$\text{TRANZ}_i = \frac{\text{CDARCY} \cdot \text{TMLTZ}_i \cdot A}{B} \quad (\text{TD 2.5})$$

$$A = \frac{DZ_i \cdot DX_j \cdot DY_j + DZ_j \cdot DX_i \cdot DY_i}{DZ_i + DZ_j} \quad (\text{TD 2.6})$$

$$B = \frac{\frac{DZ_i}{\text{PERMZ}_i} + \frac{DZ_j}{\text{PERMZ}_j}}{2} \quad (\text{TD 2.7})$$

No dip correction in Z; no net-to-gross factor either (RNTG is a horizontal-flow correction only).

3 Alternative block-centred form (OLDTRANR)

Activated by OLDTRANR keyword in the GRID section. Different combination of cell-cross-section and permeability per cell.

$$\text{TRANX}_i = \frac{\text{CDARCY} \cdot \text{TMLTX}_i \cdot \text{DIPC}}{B} \quad (\text{TD 2.8})$$

$$B = \frac{\frac{DX_i}{A_i \cdot \text{PERMX}_i} + \frac{DX_j}{A_j \cdot \text{PERMX}_j}}{2} \quad (\text{TD 2.9})$$

with $A_i = DY_i \cdot DZ_i \cdot \text{NTG}_i$ and $A_j = DY_j \cdot DZ_j \cdot \text{NTG}_j$ (per-cell cross-sections). The dip correction follows TD Eq. 2.4/(TD 2.4).

Z-direction (OLDTRANR)

$$\text{TRANZ}_i = \frac{\text{CDARCY} \cdot \text{TMLTZ}_i}{B} \quad (\text{TD 2.11})$$

$$B = \frac{\frac{DZ_i}{A_i \cdot \text{PERMZ}_i} + \frac{DZ_j}{A_j \cdot \text{PERMZ}_j}}{2} \quad (\text{TD 2.12})$$

with $A_i = DX_i \cdot DY_i$ and $A_j = DX_j \cdot DY_j$ (no NTG in vertical).

4 Corner-point transmissibility (NEWTRAN)

The workhorse formula. Handles both regular face connections and fault-displaced NNCs uniformly.

X-direction transmissibility

$$\text{TRANX}_{ij} = \frac{\text{CDARCY} \cdot \text{TMLTX}_i}{\frac{1}{T_i} + \frac{1}{T_j}} \quad (\text{TD 2.13})$$

The half-cell transmissibility for cell i is

$$T_i = \text{PERMX}_i \cdot \text{RNTG}_i \cdot \frac{\mathbf{A} \cdot \mathbf{D}_i}{\mathbf{D}_i \cdot \mathbf{D}_i} \quad (1)$$

where

$$\mathbf{A} \cdot \mathbf{D}_i = A_x DX_i + A_y DY_i + A_z DZ_i \quad (2)$$

$$\mathbf{D}_i \cdot \mathbf{D}_i = DX_i^2 + DY_i^2 + DZ_i^2 \quad (3)$$

with:

- (A_x, A_y, A_z) — projections of the mutual interface area onto the global X, Y, Z axes (computed by Eclipse from the corner-point overlap)
- (DX_i, DY_i, DZ_i) — components of the vector from cell- i centre to the centre of the face connecting cells i and j (*not* cell side lengths in NEWTRAN context)

T_j is computed analogously for cell j (using PERMX_j , RNTG_j , and $\mathbf{D}_j$ — the vector from cell- j centre to the same face centre).

Why this works for NNCs: for an axis-aligned face neighbour pair, $A_y = A_z = 0$ and the formula collapses to the half-cell harmonic mean. For an NNC across a fault, $\mathbf{A}$ encodes the partial overlap geometry (which may have all three components non-zero), and the dot-product structure correctly projects the flow path onto the permeability tensor.

Y- and Z-direction transmissibility

Y-transmissibility uses PERMY in place of PERMX. Z-transmissibility uses PERMZ *without* the RNTG factor (vertical flow is between full layer interfaces, no net-to-gross reduction).

5 Half-cell permeability scheme (HALFTRAN)

Eclipse 100. Allows six effective half-cell permeabilities per cell: K_{xil} , K_{xir} , K_{yil} , K_{yir} , K_{zil} , K_{zir} (left/right of each cell in each direction).

The half-cell transmissibilities are

$$T_{il} = K_{xil} \cdot \text{RNTG}_i \cdot \frac{\mathbf{A} \cdot \mathbf{D}_i}{\mathbf{D}_i \cdot \mathbf{D}_i} \quad (\text{TD 2.14})$$

$$T_{ir} = K_{xir} \cdot \text{RNTG}_i \cdot \frac{\mathbf{A} \cdot \mathbf{D}_i}{\mathbf{D}_i \cdot \mathbf{D}_i} \quad (\text{TD 2.15})$$

Implemented by storing $\text{MULTX}_i = K_{xir} / \text{PERMX}_i$ and $\text{MULTX-}_i = K_{xil} / \text{PERMX}_i$, with normal PERMX values entered.

Requires GRIDOPTS YES in RUNSPEC to enable the negative-direction multipliers (MULTX-, MULTY-, MULTZ-).

6 Radial transmissibility

For DR/DTHETA/DZ grids:

$$\text{TRANR}_i = \frac{\text{CDARCY} \cdot \text{TMLTR}_i \cdot \text{DIPC}}{\frac{1}{T_i} + \frac{1}{T_j}} \quad (\text{TD 2.16})$$

with pressure-equivalent radii r_1, r_2, r_3 giving

$$\begin{aligned} T_i &= \frac{\text{PERMR}_i \text{ RNTG}_i D\theta_i DZ_i}{F_{1i}} & F_{1i} &= \frac{r_1^2}{r_2^2 - r_1^2} \ln(r_1/r_2) + \frac{1}{2} \\ T_j &= \frac{\text{PERMR}_j \text{ RNTG}_j D\theta_j DZ_j}{F_{2j}} & F_{2j} &= \frac{r_3^2}{r_3^2 - r_2^2} \ln(r_3/r_2) - \frac{1}{2} \end{aligned}$$

where r_2 is the radius of the cell interface and r_1, r_3 are pressure-equivalent radii of cells i and j .

7 Non-Neighbour Connections

Eclipse auto-generates an NNC between two cells whenever they have a non-zero mutual interface area but are not Cartesian neighbours. Sources:

1. Corner-point fault offset (the canonical case — see below)
2. Pinch-out bridging (PINCH)
3. Local-grid-refinement boundary (CARFIN/RADFIN)
4. Numerical aquifer connection (AQUNUM/AQUCON)
5. Dual porosity / dual permeability matrix-fracture coupling (DUALPORO/DUALPERM)

Geometric NNC transmissibility

For cases 1–3 above, Eclipse computes the NNC transmissibility via the *same* NEWTRAN formula (TD 2.13)–(3), with the appropriate mutual interface area projections (A_x, A_y, A_z) for the partial overlap. The formula is uniform across regular faces and NNCs.

Aquifer NNC

Numerical aquifer transmissibilities use specialised formulas defined by AQUCON parameters (area, length, permeability). Beyond the scope of this document.

Dual-porosity NNC (cross-reference only)

$$T_{mf} = \text{CDARCY} \cdot \sigma \cdot K \cdot V \quad (\text{TD 2.54})$$

where σ is the Kazemi shape factor, K is the matrix permeability, V is the matrix bulk volume. See sibling library `eclipse-dual-porosity` for full derivation.

User-defined NNC transmissibility (NNC keyword)

The user enters the transmissibility directly:

$$T_{ij}^{\text{user}} = \text{TRAN} \quad (\text{item 7 of the NNC record}) \quad (4)$$

If the user also supplies an area A_{ij} (item 16) and linking permeability K_{ij}^{link} (item 17) — both required for non-Darcy flow (VDFLOW) — Eclipse back-computes the half-cell distances:

$$DX_i + DX_j = \frac{2 A_{ij} K_{ij}^{\text{link}} c_{\text{conv}}}{T_{ij}^{\text{user}}} \quad (5)$$

where c_{conv} is a unit conversion factor. Implementation split: $DX_i = DX_j = (DX_i + DX_j)/2$.

Combined multiplier semantics

For any connection between cells i and j , the final transmissibility is the product:

$$T_{\text{final}} = T_{\text{base}} \cdot M_{i,j}^{(\text{cell})} \cdot M^{\text{fault}} \cdot M^{\text{region}} \cdot M^{\text{EDITNNC}} \quad (6)$$

where

- T_{base} is the geometric Tx from NEWTRAN (TD 2.13) or OLDTRAN (TD 2.1);
- $M_{i,j}^{(\text{cell})}$ is the product of per-cell MULTX/Y/Z (and MULTX-/Y-/Z- where applicable);
- M^{fault} is the MULTFLT multiplier for the named fault face containing this connection (1 if no fault);
- M^{region} is the MULTREGT multiplier between the cells' MULTNUM regions (1 if same region);
- M^{EDITNNC} is the EDITNNC multiplier (1 if not edited; replaced by a constant if EDITNNCR is used).

8 Dip correction DIPC

For block-centred OLDTRAN/OLDTRANR:

$$\text{DIPC} = \frac{\text{DHS}}{\text{DHS} + \text{DVS}} \quad , \quad \text{DHS} = \left(\frac{DX_i + DX_j}{2} \right)^2 \quad , \quad \text{DVS} = (\text{DEPTH}_j - \text{DEPTH}_i)^2 \quad (\text{TD 2.4/2.10})$$

Limit cases: horizontal layer $\Rightarrow$ DVS = 0 $\Rightarrow$ DIPC = 1. Highly dipping $\Rightarrow$ DIPC < 1, reducing effective Tx. Applied only to X and Y (not Z). NEWTRAN does not require an explicit DIPC — the corner-point geometry already encodes dip in $\mathbf{D}_i, \mathbf{D}_j$.

9 Equation $\rightarrow$ keyword cross-reference

Equation	Concept	Controlling keyword(s)
(TD 2.1)–(TD 2.4)	Block-centred X Tx (OLDTRAN)	OLDTRAN, DX, DY, DZ, PERMX, MULTX
(TD 2.5)–(TD 2.7)	Block-centred Z Tx	OLDTRAN, DZ, PERMZ, MULTZ
(TD 2.8)–(TD 2.12)	Alternative block-centred	OLDTRANR
(TD 2.13)–(3)	Corner-point Tx (NEWTRAN, NNC formula)	NEWTRAN, COORD, ZCORN
(TD 2.14)–(TD 2.15)	Half-cell perm scheme	HALFTRAN, MULTX-, MULTY-, MULTZ-, GRIDOPTS
(TD 2.16)	Radial Tx	DR, DTHETA, INRAD
(4)–(5)	User-defined NNC	NNC, NNCGEN
(TD 2.54)	DP/DK matrix-fracture NNC	DUALPORO/DUALPERM, SIGMA, SIGMAV
(6)	Combined multiplier product	MULTX, MULTFLT, MULTREGT, EDITNNC, EDITNNCR
(TD 2.4/2.10)	Dip correction (OLDTRAN family only)	implicit; not user-controlled

10 Default CDARCY values

Unit system	E100 value	E300 value
METRIC	0.008527	0.00852702
FIELD	0.001127	0.00112712
LAB	3.6	3.6
PVT-M	0.00864	0.00864

These conversion constants give transmissibilities in units of cP·reservoir-volume / (day · pressure) — specifically: cP · rm³/day/bars (METRIC), cP · rb/day/psi (FIELD), cP · rcc/hr/atm (LAB), cP · rm³/day/atm (PVT-M).

11 NNC drop / inline thresholds

- NNCs with $T < 10^{-6}$ are silently dropped from the connection list (RM p.1468).
- NNCs with $T < 10^{-20}$ produce a warning and are dropped.
- NNCs whose path is geometrically equivalent to a regular face connection (e.g. entire intervening layer inactive) are *inlined* — converted back to regular face connections internally. Visible only via the ALLNNC mnemonic in RPTGRID.

12 Composition with the DP/DK system

For a fully coupled DP+faulted run, the Jacobian contains connections from:

1. Standard NEWTRAN/OLDTRAN face-neighbour connections (fracture-fracture and, under DK, matrix-matrix)
2. Auto-generated geometric NNCs from corner-point fault offsets — same NEWTRAN formula with partial overlap area
3. Pinch-out NNCs from PINCH
4. Dual-porosity σ -NNCs (separate Kazemi formula, (TD 2.54))
5. Block-to-block NNCs from BTOBALFA (standard spatial Tx with area multiplier)

MULTFLT applies to (1), (2), (3), and (5). It does *not* apply to (4) (DP σ -NNCs); modify those via SIGMAV or MULTSIGV.

A Equation index (by TD equation number)

TD Eq.	Label in this document
2.1	(TD 2.1)
2.2	(TD 2.2)
2.3	(TD 2.3)
2.4	(TD 2.4)
2.5	(TD 2.5)
2.6	(TD 2.6)
2.7	(TD 2.7)
2.8	(TD 2.8)
2.9	(TD 2.9)
2.10	(TD 2.4/2.10)
2.11	(TD 2.11)
2.12	(TD 2.12)
2.13	(TD 2.13)
2.14	(TD 2.14)
2.15	(TD 2.15)
2.16	(TD 2.16)

End of mathematical model.